

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12417390
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Marcu; Alon et al.

Enhanced digital signal processor (DSP) NAND flash

Abstract

A method and apparatus for systems and methods for digital signal processing (DSP) in a non-volatile memory (NVM) device comprising CMOS coupled to NVM die, of a data storage device. According to certain embodiments, one or more DSP calculations are provided by a controller to the CMOS components of the NVM, that configure one or more memory die to carry out atomic calculations on the data resident on the die. The results of calculations of each die are provided to an output latch for each die, back-propagating data back to the configured calculation portion as needed, otherwise forwarding the results to the controller. The controller aggregates the results of DSP calculations of each die and presents the results to the host system.

Inventors: Marcu; Alon (Tel-Mond, IL), Navon; Ariel (Revava, IL), Hahn; Judah Gamliel (Ofra, IL), Benisty; Shay (Beer Sheva, IL), Sharon; Eran (Rishon Lezion, IL), Inbar; Karin (Ramat Hasharon, IL)

Applicant: Sandisk Technologies, Inc. (Milpitas, CA)

Family ID: 1000008818950

Assignee: Sandisk Technologies, Inc. (Milpitas, CA)

Appl. No.: 17/349508

Filed: June 16, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220405601 A1	Dec. 22, 2022

Publication Classification

Int. Cl.: G06N3/084 (20230101); G06F7/544 (20060101); G06F7/57 (20060101); G06N3/06 (20060101)

U.S. Cl.:

CPC **G06N3/084** (20130101); **G06F7/5443** (20130101); **G06F7/57** (20130101); **G06N3/061** (20130101);

Field of Classification Search

CPC: G06N (3/084); G06N (3/061); G06N (3/063); G06F (7/5443); G06F (7/57); G06F (3/061); G06F (3/0659); G06F (3/0679); G06F (3/0688); G06F (2212/1024); G06F (2212/7203); G06F (12/0238); G06F (15/7821); G06F (2212/7208); G11C (11/54); G11C (16/0483)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6545907	12/2002	Lowrey et al.	N/A	N/A
7075841	12/2005	Resta et al.	N/A	N/A
7274586	12/2006	Choi et al.	N/A	N/A
7983069	12/2010	Hsu et al.	N/A	N/A
8199566	12/2011	Fackenthal et al.	N/A	N/A
8725935	12/2013	Huang et al.	N/A	N/A
9530491	12/2015	Uttarwar et al.	N/A	N/A
9646243	12/2016	Gokmen	N/A	N/A
9697077	12/2016	Sokolov et al.	N/A	N/A
9880760	12/2017	Inbar et al.	N/A	N/A
9997232	12/2017	Murphy	N/A	N/A
10741247	12/2019	Yeh et al.	N/A	N/A
2009/0138687	12/2008	Kang	N/A	N/A
2010/0027329	12/2009	Lee et al.	N/A	N/A
2012/0311262	12/2011	Franceschini et al.	N/A	N/A
2013/0013860	12/2012	Franceschini et al.	N/A	N/A
2014/0129753	12/2013	Schuette	710/301	G06F 13/4068
2016/0085464	12/2015	Tuers et al.	N/A	N/A
2016/0092129	12/2015	Agarwal et al.	N/A	N/A
2017/0063394	12/2016	Halbert et al.	N/A	N/A
2017/0068451	12/2016	Kenan et al.	N/A	N/A
2017/0200078	12/2016	Bichler	N/A	N/A
2017/0270403	12/2016	Zhang	N/A	N/A
2017/0337466	12/2016	Bayat et al.	N/A	N/A
2018/0052766	12/2017	Mehra	N/A	G06F 3/0661
2018/0075338	12/2017	Gokmen	N/A	N/A
2018/0075344	12/2017	Ma et al.	N/A	N/A
2018/0232508	12/2017	Kursun	N/A	N/A
2018/0330238	12/2017	Luciw et al.	N/A	N/A
2019/0043836	12/2018	Fastow	N/A	H01L 25/0657
2019/0243787	12/2018	Mittal et al.	N/A	N/A
2019/0258920	12/2018	Lie et al.	N/A	N/A
2020/0174749	12/2019	Kang et al.	N/A	N/A

2020/0184335	12/2019	Rom et al.	N/A	N/A
2020/0185027	12/2019	Rom et al.	N/A	N/A
2021/0342676	12/2020	Hoang	N/A	G06N 3/04
2022/0091775	12/2021	Shridhar	N/A	G06F 3/0659
2023/0153032	12/2022	Shin	711/154	G06F 3/0611

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106485317	12/2016	CN	N/A
107301455	12/2016	CN	N/A
20180028966	12/2017	KR	N/A
20180116094	12/2017	KR	N/A

OTHER PUBLICATIONS

Yangyang Pan et al., Using Quasi-EZ-NAND Flash Memory to Build Large-Capacity Solid-State Drives in Computing Systems, IEEE Transactions on Computers, vol. 62, No. 5, May 2013, 7 pages. cited by applicant

Kota Ando, et al., In-Memory Area-Efficient Signal Streaming Processor Design for Binary Neural Networks, 978-1-5090-6389-5/17/\$31.00 © 2017 IEEE, 4 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2019/050105, dated Nov. 3, 2020, 13 pages. cited by applicant

Hadidi et al, "Demystifying the Characteristics of 30-Stacked Memories: A Case Study for Hybrid Memory Cube," Georgia Institute of Technology, Oct. 3, 2017, 10 pages

<<https://arxiv.org/pdf/1706.02725.pdf>>. cited by applicant

Park et al, "A hybrid flash translation layer design for SLC-MLC flash memory based multibank solid state disk," Microprocessors and Microsystems, vol. 35, Issue 1, Feb. 2011, pp. 48-59

<https://doi.org/10.1016/j.micpro.2010.08.001>. cited by applicant

Chi et al., "Prime: A Novel Processing-in-memory Architecture for Neural Network Computation in ReRAM-based Main Memory", 2016 ACM SIGARCH Computer Architecture News Jun. 18, 2016 (vol. 44, No. 3, pp. 27-39). IEEE Press.

<https://seal.ece.ucsb.edu/sites/seal.ece.ucsb.edu/files/publications/prime_isca_2016.pdf>. cited by applicant

Choe et al, "Near-Data Processing for Machine Learning", 2017, International Conference on Learning Representations. 12 pages. <https://openreview.net/pdf?id=H1_EDpogx>. cited by applicant

Sullivan, John, "Merging Memory and Computation, Programmable Chip Speeds AI, Slashes Power Use", Nov. 15, 2018, 5 pages <<https://m.techxplare.com/news/2018-11-merging-memory-programmable-chip-ai.html>>. cited by applicant

Jia et al, "A Microprocessor Implemented in 65nm CMOS with Configurable and Bit-scalable Accelerator for Programmable In-memory Computing", Nov. 9, 2018, 10 pages

<<https://arxiv.org/pdf/1811.04047.pdf>>. cited by applicant

Valavi et al, "A Mixed-Signal Binarized Convolutional-Neural-Network Accelerator Integrating Dense Weight Storage and Multiplication for Reduced Data Movement", VLSI Symp. on Circuits {VLSIC}, Jun. 2018. 2 pages. <<http://www.princeton.edu/~nvermaNermaLabSite/Publications/2018NalaviRamadgeNestlerVermaVLS118.pdf>>. cited by applicant

Nielsen, Michael "Neural Networks and Deep Learning, Chapter 2: How the backpropagation algorithm works", 2015, 27 pages <<http://neuralnetworksanddeeplearning.com/chap2.html>>. cited by applicant

Nielsen, Michael "Neural Networks and Deep Learning, Chapter 2: How the backpropagation algorithm works", 2015, 27 pages <<http://neuralnetworksanddeeplearning.com/chap2.html>>. cited by applicant

Primary Examiner: Abad; Farley

Attorney, Agent or Firm: Patterson + Sheridan, LLP

Background/Summary

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

(1) Embodiments of the present disclosure generally relate to computation on a data storage device, and more particularly, to digital signal processing on a non-volatile memory (NVM) device.

Description of the Related Art

(2) Digital signal processing (DSP) with large amounts of data conventionally is a resource intensive task, typically requiring management of a host system to implement calculation methods, in addition to the acquisition of data from storage and processing of that data. Host system resources utilized for this task may be unavailable for other processing tasks, possibly for extended periods of time. Examples of DSP computation tasks include search, data filtering, machine learning (e.g., neural network, regression, state vector machines, clustering, and others, in addition to optimizing associated parameters), counting within large bodies of data, and the like.

(3) Therefore, there is a need in the art to efficiently execute DSP within a data storage device.

SUMMARY OF THE DISCLOSURE

(4) A method and apparatus for systems and methods for digital signal processing (DSP) in a non-volatile memory (NVM) device comprising CMOS coupled to NVM die, of a data storage device. According to certain embodiments, one or more DSP calculations are provided by a controller to the CMOS components of the NVM which configure one or more memory die to carry out atomic calculations on the data resident on the die. The results of calculations of each die are provided to an output latch for each die, back-propagating data back to the configured calculation portion as needed, otherwise forwarding the results to the controller. The controller aggregates the results of DSP calculations of each die and presents the results to the host system.

(5) In one embodiment, a data storage device is disclosed that includes a memory device comprising a plurality of CMOS structures, each CMOS structure coupled to a portion of an NVM die, and a controller. The controller is configured to cause the data storage device to receive a command to perform a digital signal processing (DSP) computation, encode a portion of the DSP computation to each CMOS structure of the plurality of CMOS structures of the memory device; and receive an output of the DSP computation from the plurality of CMOS structures of the memory device.

(6) In another embodiment, a controller for a data storage device is disclosed, that includes a memory device comprising computer-readable instructions, and a processor configured to read the computer-readable instructions that cause the processor to execute a method for digital signal processing. The method includes storing a portion of data of a data set on an NVM die of a plurality of NVM dies, the portion of data comprising an atomic unit of data, receiving a digital signal processing (DSP) command comprising a plurality of calculations, and encoding a CMOS of an NVM memory device comprising the CMOS coupled to an NVM die of the plurality of NVM dies, with a calculation of the plurality of calculations. The method further includes processing the calculation on the NVM die with the portion of data stored on the NVM die, and receiving a result of processing the calculation.

(7) In another embodiment, a data storage device is disclosed, that includes a memory means, and a controller means configured to perform a method for digital signal processing (DSP). The method includes receiving a command to perform a DSP calculation comprising a plurality of atomic calculation components, encoding a plurality of dies of the memory means, each die encoded to calculate at least one of the plurality of atomic calculation components, and receiving an output of the DSP calculation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

(2) FIG. 1 is a schematic block diagram illustrating a storage system in which a data storage device may function as a storage device for a host device, according to certain embodiments.

(3) FIG. 2 is a schematic illustration of a Circuit Bounded Array (CbA) architecture, according to certain embodiments.

(4) FIG. 3 is a schematic block diagram illustrating a controller managing a plurality of memory dies each including a CMOS chip of an NVM, according to certain embodiments.

(5) FIG. 4 is a schematic block diagram illustrating a CMOS chip architecture, according to certain embodiments.

(6) FIG. 5 is a flow diagram illustrating a method of digital signal processing, according to certain embodiments.

(7) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.

DETAILED DESCRIPTION

(8) In the following, reference is made to embodiments of the disclosure. However, it should be understood that the disclosure is not limited to specifically described embodiments. Instead, any combination of the following features and elements, whether related to different embodiments or not, is contemplated to implement and practice the disclosure. Furthermore, although embodiments of the disclosure may achieve advantages over other possible solutions and/or over the prior art, whether or not a particular advantage is achieved by a given embodiment is not limiting of the disclosure. Thus, the following aspects, features, embodiments, and advantages are merely illustrative and are not considered elements or limitations of the appended claims except where explicitly recited in a claim(s). Likewise, reference to “the disclosure” shall not be construed as a generalization of any inventive subject matter disclosed herein and shall not be considered to be an element or limitation of the appended claims except where explicitly recited in a claim(s).

(9) A method and apparatus for systems and methods for digital signal processing (DSP) in a non-volatile memory (NVM) device comprising CMOS coupled to NVM die, of a data storage device. According to certain embodiments, one or more DSP calculations are provided by a controller to the CMOS components of the NVM, that configure one or more memory die to carry out atomic calculations on the data resident on the die. The results of calculations of each die are provided to an output latch for each die, back-propagating data back to the configured calculation portion as needed, otherwise forwarding the results to the controller. The controller aggregates the results of

DSP calculations of each die and presents the results to the host system.

(10) FIG. 1 is a schematic block diagram illustrating a storage system **100** in which a host device **104** is in communication with a data storage device **106**, according to certain embodiments. For instance, the host device **104** may utilize a non-volatile memory (NVM) **110** included in data storage device **106** to store and retrieve data. The host device **104** comprises a host DRAM **138**. In some examples, the storage system **100** may include a plurality of storage devices, such as the data storage device **106**, which may operate as a storage array. For instance, the storage system **100** may include a plurality of data storage devices **106** configured as a redundant array of inexpensive/independent disks (RAID) that collectively function as a mass storage device for the host device **104**.

(11) The host device **104** may store and/or retrieve data to and/or from one or more storage devices, such as the data storage device **106**. As illustrated in FIG. 1, the host device **104** may communicate with the data storage device **106** via an interface **114**. The host device **104** may comprise any of a wide range of devices, including computer servers, network attached storage (NAS) units, desktop computers, notebook (i.e., laptop) computers, tablet computers, set-top boxes, telephone handsets such as so-called “smart” phones, so-called “smart” pads, televisions, cameras, display devices, digital media players, video gaming consoles, video streaming device, or other devices capable of sending or receiving data from a data storage device.

(12) The data storage device **106** includes a controller **108**, NVM **110**, a power supply **111**, volatile memory **112**, the interface **114**, and a write buffer **116**. In some examples, the data storage device **106** may include additional components not shown in FIG. 1 for the sake of clarity. For example, the data storage device **106** may include a printed circuit board (PCB) to which components of the data storage device **106** are mechanically attached and which includes electrically conductive traces that electrically interconnect components of the data storage device **106** or the like. In some examples, the physical dimensions and connector configurations of the data storage device **106** may conform to one or more standard form factors. Some example standard form factors include, but are not limited to, 3.5" data storage device (e.g., an HDD or SSD), 2.5" data storage device, 1.8" data storage device, peripheral component interconnect (PCI), PCI-extended (PCI-X), PCI Express (PCIe) (e.g., PCIe x1, x4, x8, x16, PCIe Mini Card, MiniPCI, etc.). In some examples, the data storage device **106** may be directly coupled (e.g., directly soldered or plugged into a connector) to a motherboard of the host device **104**.

(13) Interface **114** may include one or both of a data bus for exchanging data with the host device **104** and a control bus for exchanging commands with the host device **104**. Interface **114** may operate in accordance with any suitable protocol. For example, the interface **114** may operate in accordance with one or more of the following protocols: advanced technology attachment (ATA) (e.g., serial-ATA (SATA) and parallel-ATA (PATA)), Fibre Channel Protocol (FCP), small computer system interface (SCSI), serially attached SCSI (SAS), PCI, and PCIe, non-volatile memory express (NVMe), OpenCAPI, GenZ, Cache Coherent Interface Accelerator (CCIX), Open Channel SSD (OCSSD), or the like. Interface **114** (e.g., the data bus, the control bus, or both) is electrically connected to the controller **108**, providing an electrical connection between the host device **104** and the controller **108**, allowing data to be exchanged between the host device **104** and the controller **108**. In some examples, the electrical connection of interface **114** may also permit the data storage device **106** to receive power from the host device **104**. For example, as illustrated in FIG. 1, the power supply **111** may receive power from the host device **104** via interface **114**.

(14) The NVM **110** may include a plurality of memory devices or memory units. NVM **110** may be configured to store and/or retrieve data. For instance, a memory unit of NVM **110** may receive data and a message from controller **108** that instructs the memory unit to store the data. Similarly, the memory unit may receive a message from controller **108** that instructs the memory unit to retrieve data. In some examples, each of the memory units may be referred to as a die. In some examples, the NVM **110** may include a plurality of dies (i.e., a plurality of memory units). In some examples,

each memory unit may be configured to store relatively large amounts of data (e.g., 128 MB, 256 MB, 512 MB, 1 GB, 2 GB, 4 GB, 8 GB, 16 GB, 32 GB, 64 GB, 128 GB, 256 GB, 512 GB, 1 TB, etc.).

(15) In some examples, each memory unit may include any type of non-volatile memory devices, such as flash memory devices, phase-change memory (PCM) devices, resistive random-access memory (ReRAM) devices, magneto-resistive random-access memory (MRAM) devices, ferroelectric random-access memory (F-RAM), holographic memory devices, and any other type of non-volatile memory devices.

(16) The NVM **110** may comprise a plurality of flash memory devices or memory units. NVM Flash memory devices may include NAND or NOR based flash memory devices and may store data based on a charge contained in a floating gate of a transistor for each flash memory cell. In NVM flash memory devices, the flash memory device may be divided into a plurality of dies, where each die of the plurality of dies includes a plurality of physical or logical blocks, which may be further divided into a plurality of pages. Each block of the plurality of blocks within a particular memory device may include a plurality of NVM cells. Rows of NVM cells may be electrically connected using a word line to define a page of a plurality of pages. Respective cells in each of the plurality of pages may be electrically connected to respective bit lines. Furthermore, NVM flash memory devices may be 2D or 3D devices and may be single level cell (SLC), multi-level cell (MLC), triple level cell (TLC), or quad level cell (QLC). The controller **108** may write data to and read data from NVM flash memory devices at the page level and erase data from NVM flash memory devices at the block level.

(17) The power supply **111** may provide power to one or more components of the data storage device **106**. When operating in a standard mode, the power supply **111** may provide power to one or more components using power provided by an external device, such as the host device **104**. For instance, the power supply **111** may provide power to the one or more components using power received from the host device **104** via interface **114**. In some examples, the power supply **111** may include one or more power storage components configured to provide power to the one or more components when operating in a shutdown mode, such as where power ceases to be received from the external device. In this way, the power supply **111** may function as an onboard backup power source. Some examples of the one or more power storage components include, but are not limited to, capacitors, super-capacitors, batteries, and the like. In some examples, the amount of power that may be stored by the one or more power storage components may be a function of the cost and/or the size (e.g., area/volume) of the one or more power storage components. In other words, as the amount of power stored by the one or more power storage components increases, the cost and/or the size of the one or more power storage components also increases.

(18) The volatile memory **112** may be used by controller **108** to store information. Volatile memory **112** may include one or more volatile memory devices. In some examples, controller **108** may use volatile memory **112** as a cache. For instance, controller **108** may store cached information in volatile memory **112** until the cached information is written to the NVM **110**. As illustrated in FIG. **1**, volatile memory **112** may consume power received from the power supply **111**. Examples of volatile memory **112** include, but are not limited to, random-access memory (RAM), dynamic random access memory (DRAM), static RAM (SRAM), and synchronous dynamic RAM (SDRAM (e.g., DDR1, DDR2, DDR3, DDR3L, LPDDR3, DDR4, LPDDR4, and the like)).

(19) Controller **108** may manage one or more operations of the data storage device **106**. For instance, controller **108** may manage the reading of data from and/or the writing of data to the NVM **110**. In some embodiments, when the data storage device **106** receives a write command from the host device **104**, the controller **108** may initiate a data storage command to store data to the NVM **110** and monitor the progress of the data storage command. Controller **108** may determine at least one operational characteristic of the storage system **100** and store the at least one operational characteristic in the NVM **110**. In some embodiments, when the data storage device

106 receives a write command from the host device **104**, the controller **108** temporarily stores the data associated with the write command in the internal memory or write buffer **116** before sending the data to the NVM **110**.

(20) FIG. 2 is a schematic illustration of a Circuit Bounded Array (CbA) architecture **200**, according to one embodiment. The CbA architecture **200** includes one or more complementary metal oxide semiconductor (CMOS) chips **206** coupled to respective one or more memory arrays **208** via one or more connection units. In one embodiment, each memory array **208** comprises a memory die. The CMOS chips **206** may include an encoder unit and a decoder unit as well as similar processing logic as a controller, such as the controller **108**. The CMOS chips **206** may further include an error correction code (ECC) engine, which may be part of the encoder unit and the decoder unit. The CMOS chips **206** may be configured to schedule received write commands to program data to the coupled memory die and schedule received read commands to read data from the coupled memory die. It is to be understood that that NAND dies may be referred to as memory dies for exemplary purposes.

(21) In the description herein, it is to be understood that CMOS chips and CbA architecture may be used interchangeably for exemplary purposes, and embodiments described herein may also be applicable to CMOS Under Array (CuA) architectures and CMOS Above Array (CAA) architectures.

(22) FIG. 3 is a schematic block diagram **300** illustrating a controller **302** managing a plurality of memory dies **306a-306n** each, including a CMOS chip of an NVM stack **304**, according to certain embodiments. The NVM stack **304** may be the NVM **110** of FIG. 1. The controller **302** may be a controller of a host device, such as the host device **104** of FIG. 1, or a controller of the data storage device, such as the controller **108** of FIG. 1. Each CMOS chip, such as the CMOS chip **206** of FIG. 2, of each memory die of the plurality of memory dies **306a-306n** may be configured to handle the computational tasks and other relevant data management operations. Furthermore, the CMOS chips may allow for enhanced parallel execution of DSP operations at the NVM level (e.g., at the memory die level).

(23) Parallel computation operations include reading data from the NVM stacks **304** and running operations, such as filtering or searching contents, counting repeated strings, etc. Such operations may require reading large chunks of data from the NVM stack **304**, executing logic, such as computational logic, and sending results of the computation logic that may include smaller volumes of data back to the controller **302**. Thus, the amount of data transferred to the controller **302** may be reduced, power may be saved, and performance may be improved. The controller **302** may activate one or more memory dies of the plurality of memory dies **306a-306n** in parallel to execute the logic with each memory die operating on the data that is written in its own memory. The results of the logic are gathered by the controller **302**.

(24) Another type of computational operation may involve of reading data from the NVM stack **304**, performing an operation (e.g., compression, image analysis, and filtering, or a mathematical transformation) on the read data, and writing the result of the operation back to the respective memory die of the plurality of memory dies **306a-306n**. The data that is operated on may be contained within the memory die and the respective CMOS chip of the memory die. For example, rather than transferring the data to the controller **302** to be operated on, the respective CMOS chip may include the same logic to operate on the data. Thus, the data remains closer to the respective memory die. In one example, the controller **302** may ensure that data that requires internal computational processing or transformation may be processed in parallel by the respective CMOS chips of the plurality of memory dies **306a-306n** without transfer between the memory dies.

(25) FIG. 4 is a schematic block diagram illustrating a CMOS chip architecture **400**, according to certain embodiments. The CMOS chip architecture **400** includes a CMOS chip **402**. In some examples, the CMOS chip **402** may be the CMOS chip **206** of FIG. 2. The CMOS chip **402** includes a NVM input latch **406**, a plurality of multi-accumulate (MAC) units **408a-408n**, an NVM

output latch **410**, a central processing unit (CPU) **412**, an SRAM (**414**), and a host interface module (HIM) **416**. The CMOS chip **402** is coupled to a respective memory die of a plurality of memory dies, such as the plurality of memory dies **306a-306n** of FIG. **3**. For example, the CMOS chip **402** may be coupled to a first memory die **306a**. The NVM input latch **406** and the NVM output latch **410** may attach directly to the NVM array, such as the one or more memory arrays **208** of FIG. **2**.

(26) The CPU **412** may be configured to schedule data to be programmed to the respective memory die, access the respective memory die, and/or schedule data to be read from the respective memory die using the NVM input latch **406**. The NVM input latch **406** retrieves data from the respective memory die and transfers the data to one or more of the plurality of MAC units **408a-408n**. The plurality of MAC units **408a-408n** processes the input data from the NVM input latch **406** in parallel, such that parallel DSP functions may be performed. It is to be understood that the plurality of MAC units **408a-408n** may be one or more MAC units, such as about 64, 128, 256, 512, 1024, and more MAC units. It is to be understood that the plurality MAC units **408a-408n** may alternatively be other computational modules, or other computational modules may be included with the plurality of MAC units **408a-408n**. The other computational modules may include error correction code (ECC) engines, arithmetic logic units (ALU), CPUs, and the like.

(27) The results of the plurality of MAC units **408a-408n** are accumulated or aggregated at the NVM output latch **410**, where the results may be programmed back or back-propagated to the respective memory die via the NVM input latch **406**. The number of computations by the plurality of MAC units **408a-408n**, or N filters, may be completed in parallel. For example, when N equals 512, 1024 computations may be completed in parallel. Additional functionality may be implemented by CPU **412** by using the SRAM **414**. Furthermore, the HIM **416** is an interface that allows the host or a controller, such as the controller **108** of FIG. **1**, to access and manage the parallel DSP functions. Referring to FIG. **1**, the HIM **416** may be a direct connection between the host device **104** and the NVM **110** or a direct connection between the controller **108** and the NVM **110**. The host managed DSP functions may allow for hosts to execute intensive parallel computations, such as complex mathematical operations, filtering, counting, compression, image processing (Fourier transforms and the like), machine learning operation (such as forming a machine learning neuron), database searches, mathematical transformations, and the like.

(28) Data may be stored in the memory die according to a zoned namespace, a key value pair, a stream, an endurance group/NVM set, or a device-internal file system of the controller **108**. In one example, the CMOS chip **402** may be used with zone namespaces (ZNS) protocol, where a zone is generally targeted at a small number of die-blocks and may be reduced to a single die-block. The data included within a zone, where the zone includes one die, may be considered an atomic unit for internal processing.

(29) In another example, the CMOS chip may be used for key value (KV) pairs, where the KV interfaces assume an internal device processing for assigning a database structure. The KV pairs may be optimally placed in discrete dies for search and update processing. In yet another example, data segregation may be used with streams in order to isolate segments that should fully be contained within a DSP memory die for processing. The DSP memory die may refer to a memory die that is coupled to the CMOS chip **402**. Furthermore, the streams may also be used to mark which data units may be placed in conventional non-volatile storage side-by-side with the DSP memory dies.

(30) In yet another example, endurance groups/NVM sets may be used with the CMOS chip **402**. The endurance groups/NVM sets may allow for die segregation which isolates specific dies or die groups that may be used for local DSP processing, while allowing other die groups to be used for general purpose storage that does not require the DSP processing. In yet another example, the CMOS chip **402** may be used to present a file system that can be remotely mounted in a network interface. The files may be discrete objects, and the file system abstraction may include attributes that would indicate whether a specific file should be stored in a manner that allows post-processing

using the DSP memory dies. Furthermore, post-processing algorithms may be utilized for the DSP memory dies.

(31) Thus, the CMOS chip **402** may allow for a host managed DSP in the memory device, such as the NVM **110** of FIG. **1**. Thus, the host, such as the host device **104** of FIG. **1** or the controller **108** of FIG. **1**, may have the ability to perform DSP and ECC operations in parallel in memory while avoiding unnecessary data transfers between the memory dies and the host. Thus, the inputs and the results of the computations may be transferred from the host to the memory die.

(32) FIG. **5** is a flow diagram illustrating a method **500** of digital signal processing (DSP), according to certain embodiments. Method **500** may be implemented by the CMOS chip **402** of FIG. **4**. At block **502**, a portion of the data of a data set is stored on an NVM die of a plurality of NVM dies. The NVM die may be one of the plurality of memory dies **306a-306n** of FIG. **3**. At block **504**, a controller, such as the controller **108** of FIG. **1**, receives a DSP command. The DSP command includes a plurality of calculations, which may include forming a machine learning neuron, a mathematical transformation, image analysis, counting, searching, filtering, and the like. At block **506**, a CMOS chip of the NVM memory device is encoded. The encoding includes encoding at least a portion of the data of the NVM die with the calculations of the DSP command. The encoding may include using a MAC unit, such as one of the plurality of MAC units **408a-408n** of FIG. **4**, an ALU, or a CPU. The encoding may occur on separate units in parallel.

(33) At block **508**, the calculations are processed on the NVM die with the portion of the data stored on the NVM die. The calculations may be completed in parallel using the plurality of MAC units **408a-408n**. At block **510**, the result of processing the calculations are received at an output latch, such as the NVM output latch **410** of FIG. **4**, of the CMOS chip **402**. The computational results from each of the plurality of MAC units **408a-408n** are aggregated at the NVM output latch **410**. The computational results may be programmed back to the respective NVM memory device or NVM die.

(34) By using a host managed DSP with parallel computational capabilities, intensive computations may be completed in the memory device without a need to transfer intermediate data between the controller and the memory dies, thus improving latency and data storage device performance.

(35) In one embodiment, a data storage device is disclosed that includes a memory device comprising a plurality of CMOS structures, each CMOS structure coupled to a portion of an NVM die, and a controller. The controller is configured to cause the data storage device to receive a command to perform a digital signal processing (DSP) computation, encode a portion of the DSP computation to each CMOS structure of the plurality of CMOS structures of the memory device; and receive an output of the DSP computation from the plurality of CMOS structures of the memory device.

(36) The CMOS encodes a portion of the NVM to calculate the portion of the DSP computation. The NVM calculates the portion of the DSP computation using data stored on the NVM. The CMOS encodes a portion of the NVM as one of a multiply-accumulate unit (MAC), an arithmetic logic unit (ALU), or a central processing unit (CPU). The DSP computation comprises one of filtering, searching, counting, compression, image analysis, a mathematical transformation, or forming a machine learning neuron. The data storage device further includes a second memory device. The controller is further configured to encode a portion of the DSP computation to the second memory device and receive a second output of the DSP computation from the second memory device. The controller is further configured to aggregate the output and second output in response to the command.

(37) In another embodiment, a controller for a data storage device is disclosed, that includes a memory device comprising computer-readable instructions, and a processor configured to read the computer-readable instructions that cause the processor to execute a method for digital signal processing. The method includes storing a portion of data of a data set on an NVM die of a plurality of NVM dies, the portion of data comprising an atomic unit of data, receiving a digital

signal processing (DSP) command comprising a plurality of calculations, and encoding a CMOS of an NVM memory device comprising the CMOS coupled to an NVM die of the plurality of NVM dies, with a calculation of the plurality of calculations. The method further includes processing the calculation on the NVM die with the portion of data stored on the NVM die, and receiving a result of processing the calculation.

(38) The method further includes storing a second portion of data of the data set on a second NVM die of the plurality of NVM dies and processing the calculation on the second NVM die. The second NVM die is coupled to the CMOS. The receiving of processing the calculation includes receiving a result of processing the calculation of the second NVM die. The CMOS encodes a portion of the NVM die with the calculation, comprising one of a multiply-accumulate unit (MAC), an arithmetic logic unit (ALU), or a central processing unit (CPU). The DSP command comprises one of forming a machine learning neuron, a mathematical transformation, image analysis, counting, searching, or filtering. The storing the portion of data of the data set comprises storing according to one of a zoned namespace, a key value pair, a stream, an endurance group/NVM set, and a device-internal file system of the controller. The method further comprising providing the result of processing the calculation to an output latch of the NVM. The controller further comprising back-propagating a portion of the result of processing the calculation to the NVM.

(39) In another embodiment, a data storage device is disclosed, that includes a memory means, and a controller means configured to perform a method for digital signal processing (DSP). The method includes receiving a command to perform a DSP calculation comprising a plurality of atomic calculation components, encoding a plurality of dies of the memory means, each die encoded to calculate at least one of the plurality of atomic calculation components, and receiving an output of the DSP calculation.

(40) The method further comprising the controller means aggregating output of each encoded die. The encoding the plurality of die of the memory means comprises encoding at least one of the plurality of die to function as a multiply-accumulate unit (MAC), an arithmetic logic unit, or a central processing unit (CPU). The DSP calculation comprises one of filtering, searching, counting, compression, image analysis, mathematical transformation, or machine learning. The method further comprising providing the output of the DSP calculation for one of the plurality of die of the memory means to an output latch, and back-propagating contents of the output latch back to the one of the plurality of die of the memory means.

(41) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A data storage device, comprising: a memory device comprising a plurality of CMOS structures, each CMOS structure coupled to a portion of an NVM die; and a controller configured to cause the data storage device to: receive a command to perform a digital signal processing (DSP) computation; encode a portion of the DSP computation to each CMOS structure of the plurality of CMOS structures of the memory device; and receive an output of the DSP computation from the plurality of CMOS structures of the memory device.

2. The data storage device of claim 1, wherein the CMOS structure encodes a portion of the NVM to calculate the portion of the DSP computation, and wherein the encoding by one or more CMOS structures of the plurality of CMOS structures occurs in parallel.

3. The data storage device of claim 2, wherein the NVM calculates the portion of the DSP computation, using data stored on the NVM.

4. The data storage device of claim 3, wherein the CMOS structure encodes a portion of the NVM as one of a multiply-accumulate unit (MAC), an arithmetic logic unit (ALU), or a central

processing unit (CPU).

5. The data storage device of claim 4, wherein the DSP computation comprises one of filtering, searching, counting, compression, image analysis, a mathematical transformation, or forming a machine learning neuron.

6. The data storage device of claim 5, comprising a second memory device, the controller further configured to encode a portion of the DSP computation to the second memory device and receive a second output of the DSP computation from the second memory device.

7. The data storage device of claim 6, the controller further configured to aggregate the output and second output in response to the command.

8. A controller for a data storage device, comprising: a memory device comprising computer-readable instructions; and a processor configured to read the computer-readable instructions that cause the processor to execute a method for digital signal processing, the method comprising: storing a portion of data of a data set on an NVM die of a plurality of NVM dies, the portion of data comprising an atomic unit of data; receiving a digital signal processing (DSP) command comprising a plurality of calculations; encoding a CMOS of an NVM memory device comprising the CMOS coupled to an NVM die of the plurality of NVM dies, with a calculation of the plurality of calculations; processing the calculation on the NVM die with the portion of data stored on the NVM die; and receiving a result of processing the calculation.

9. The controller of claim 8, the method further comprising: storing a second portion of data of the data set on a second NVM die of the plurality of NVM dies, the second NVM die coupled to the CMOS; and processing the calculation on the second NVM die.

10. The controller of claim 9, wherein the processing the calculation includes receiving a result of processing the calculation of the second NVM die.

11. The controller of claim 10, wherein the CMOS encodes a portion of the NVM die with the calculation, comprising one of a multiply-accumulate unit (MAC), an arithmetic logic unit (ALU), or a central processing unit (CPU).

12. The controller of claim 11, wherein the DSP command comprises one of forming a machine learning neuron, a mathematical transformation, image analysis, counting, searching, or filtering.

13. The controller of claim 8, wherein storing the portion of data of the data set comprises storing according to one of a zoned namespace, a key value pair, a stream, an endurance group/NVM set, and a device-internal file system of the controller.

14. The controller of claim 8, further comprising back-propagating a portion of the result of processing the calculation to the NVM.

15. The controller of claim 10, the method further comprising providing the result of processing the calculation to an output latch of the NVM.

16. A data storage device comprising: a memory means; and a controller means configured to perform a method for digital signal processing (DSP) comprising: receiving a command to perform a DSP calculation comprising a plurality of atomic calculation components; encoding a plurality of dies of the memory means, each die encoded to calculate at least one of the plurality of atomic calculation components; and receiving an output of the DSP calculation.

17. The data storage device of claim 16, the method further comprising the controller means aggregating output of each encoded die.

18. The data storage device of claim 17, wherein encoding the plurality of die of the memory means comprises encoding at least one of the plurality of die to function as a multiply-accumulate unit (MAC), an arithmetic logic unit, or a central processing unit (CPU).

19. The data storage device of claim 18, wherein the DSP calculation comprises one of filtering, searching, counting, compression, image analysis, mathematical transformation, or machine learning.

20. The data storage device of claim 19, the method further comprising providing the output of the DSP calculation for one of the plurality of die of the memory means to an output latch, and back-

propagating contents of the output latch back to the one of the plurality of die of the memory means.
